

AUGUST/RESIT EXAMINATIONS 2022/2023

MODULE: CA318 - Advanced Algorithms and A.I. Search

PROGRAMME(S):

CASE BSc in Computer Applications (Sft.Eng.)
ECSAO Study Abroad (Engineering & Computing)

YEAR OF STUDY: 3,O

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Detailed instructions inside.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

CA318 - Advanced Algorithms and AI Search - 2023

Instructions:

Answer all 20 multiple-choice questions. (2 marks each.) Total 40 marks.

Answer 2 long questions out of 3. (25 marks each.) Total 50 marks.

Answer the bonus question. (10 marks.)

Remember to hand up the multiple-choice answer sheet!

Multiple choice

1. In machine evolution with asexual reproduction, mutation might break the parent and make it unfit. So therefore:
 1. The original statement is not true.
 2. We should set an appropriate mutation rate for the problem.
 3. We should not mutate.
 4. We should only mutate about 1 in a million bits.
 5. It is not something to worry about.
 6. We should mutate the fit, not the unfit.
 7. We should only mutate at points that will improve the parent.
 8. We should mutate the unfit, not the fit.

2. Consider a heuristic for human mate choice:

Have a relationship with n partners of fitness $> f$
Marry the next partner you find of fitness $> g$

where fitness is ranked by the individual out of 10.

Which of the following values for n, f, g (respectively) makes the most sense?

1. 100, 5, 8
 2. 100, 0, 8
 3. 10, 8, 3
 4. 10, 0, 8
 5. 100, 8, 3
 6. 10, 5, 8
3. In machine evolution, our chromosome is a binary string of length n . We interpret it as encoding an integer x . To convert this to encoding a real number in the interval a to b , we calculate:
1. $a + x(b-a)$
 2. $x/(2^n)$
 3. $a + x / (b-a)$
 4. $x/(2^n-1)$
 5. $x / (b-a)$
 6. $a + (x/(2^n-1))(b-a)$
4. The following is a correct ordering of AI methods, from lesser to greater autonomy:
1. Genetic Algorithms, Reinforcement Learning, Supervised Learning.
 2. Supervised Learning, Reinforcement Learning, Genetic Algorithms.
 3. Supervised Learning, Genetic Algorithms, Reinforcement Learning.
 4. Reinforcement Learning, Supervised Learning, Genetic Algorithms.
 5. Genetic Algorithms, Supervised Learning, Reinforcement Learning.
 6. Reinforcement Learning, Genetic Algorithms, Supervised Learning.
5. In machine evolution, our chromosome is a binary string of length n . We interpret it as encoding an integer x . To convert this to encoding a real number in the interval 0 to 1, we calculate:
1. $x/(2^{(n-1)})$
 2. $x/(2^{(n-1)}-1)$
 3. $x/(2^n)$
 4. $x/(2^n-1)$

6. Consider the parallelogram whose boundaries are these 4 lines:
 $y=0$, $y=3$, $x=y$ and $(x-3)=y$
 A multi-layer neural network architecture to fire only for points *outside* the parallelogram is:
1. 4 hidden units fire for $y \leq 0$, $y \geq 3$, $x \leq y$, and $(x-3) \geq y$, respectively, then feed their outputs into an AND gate.
 2. 4 hidden units fire for $y \leq 0$, $y \geq 3$, $x \leq y$, and $(x-3) \geq y$, respectively, then feed their outputs into an AND gate, then feed the output of that into a NOT gate.
 3. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an AND gate.
 4. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an OR gate.
 5. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an OR gate, then feed the output of that into a NOT gate.
 6. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an AND gate, then feed the output of that into a NOT gate.
7. We have a neural network where inputs are finite (positive or negative). If a node has a (negative) infinite threshold, then:
1. If weights are negative, you have constant output 1.
 2. If weights are negative, you have constant output 0.
 3. If weights are finite, you have constant output 0.
 4. None of the above.
 5. If weights are positive, you have constant output 1.
 6. If weights are finite, you have constant output 1.
 7. If weights are positive, you have constant output 0.
8. Why would we in practice *not* learn the function $y = \sin(x) + (1/x)$ from exemplars?
1. Learning it from exemplars is impossible for some values of x .
 2. It is a chaotic function.
 3. Divide by zero error.
 4. Learning it from exemplars is impossible.
 5. We *would* learn it from exemplars.
 6. It would take too long.
 7. There is nothing to learn. The problem is solved.

9. In neural networks, we need thresholds. Otherwise if all inputs are 0, then output of every single hidden node is:
1. -infinity
 2. infinity
 3. $1/2$
 4. 1
 5. 0
10. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. Which of the following is the case:
1. Neural Networks make a directed move, GAs make a directed move.
 2. Neural Networks make a random move, GAs make a directed move.
 3. Neural Networks make a random move, GAs make a random move.
 4. Neural Networks make a directed move, GAs make a random move.
11. In AI search, the following heuristic:

```
Initialise with 10 random solutions.  
Repeat forever  
{  
  Pick the best solution.  
  Make 10 new solutions by making small or no changes to it.  
}
```

might be a bad heuristic because:

1. It tends to lose good solutions once found.
2. It is not a bad heuristic.
3. It tends to converge on local optima.
4. It is not a heuristic.
5. 10 is too large.
6. It tends to converge on global optima.
7. Small changes may break the best solution.
8. 10 is too small.

12. In a multi-layer neural network, we have n hidden units with binary outputs, 1 output unit, weights $w_1, \dots, w_n$ connecting them, and threshold t for the output unit. Which one of the following implements an output unit that fires if and only if all the hidden units fire.
1. $w_1 = w_2 = \dots = w_n = 1, \quad t = 0$
 2. $w_1 = w_2 = \dots = w_n = W, \quad W > 0, \quad t = W$
 3. $w_1 = w_2 = \dots = w_n = W, \quad W > 0, \quad t = W + n$
 4. $w_1 = w_2 = \dots = w_n = W, \quad W > 0, \quad t = nW$
 5. $w_1 = w_2 = \dots = w_n = 0, \quad t = 1$
 6. $w_1 = w_2 = \dots = w_n = W, \quad W > 0, \quad t = n$
13. An upper bound on the *game-tree complexity* of Xs and Os is:
1. 3^9
 2. 3^3
 3. 9^9
 4. $9!$
 5. $3!$
 6. 9^3
14. We are learning to represent a function $Q(x)$. The input x is a 100 dimensional vector. Each dimension is an integer from 1 to 10. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:
1. 10
 2. 100
 3. 1010
 4. 110
 5. 10^{100}
 6. 1000
 7. 100^{10}

15. We are learning to represent a function $Q(x)$. The input x is a 10 dimensional vector. Each dimension is an integer from 1 to 100. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:

1. 100
2. 110
3. 1010
4. 10^{100}
5. 100^{10}
6. 1000
7. 10

16. In trying to predict boxing matches by matching to the nearest exemplar, we have two exemplars:

Opponent height - 186 cm. Opponent weight - 90 kg. Result - I won.

Opponent height - 180 cm. Opponent weight - 95 kg. Result - I lost.

The question is:

Opponent height - 185 cm. Opponent weight - 94 kg. Will I win?

We predict by *Hamming distance* to the nearest exemplar, (a) with height in cm, (b) with height in m. What does it predict?

1. (a) I win, (b) I lose.
2. (a) I lose, (b) I lose.
3. (a) I win, (b) I win.
4. (a) I lose, (b) I win.

17. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. We consider E , the error in our solution, and $\frac{\delta E}{\delta w}$ the direction in which to change our solution. Which of the

following is the case:

1. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, neither E nor

$\frac{\delta E}{\delta w}$ are known.

2. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known but

$\frac{\delta E}{\delta w}$ is unknown.

3. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known

and $\frac{\delta E}{\delta w}$ is known.

4. For Neural Networks, E is known but $\frac{\delta E}{\delta w}$ is unknown. For GAs, neither E nor

$\frac{\delta E}{\delta w}$ are known.

18. We have a neural network where inputs are finite (positive or negative). If a node has a (positive) infinite threshold, then:

1. If weights are positive, you have constant output 1.
2. If weights are positive, you have constant output 0.
3. None of the above.
4. If weights are negative, you have constant output 0.
5. If weights are finite, you have constant output 0.
6. If weights are finite, you have constant output 1.
7. If weights are negative, you have constant output 1.

19. Consider a heuristic for human mate choice:

Have a relationship with n partners of fitness $> f$
Marry the next partner you find of fitness $> g$

where fitness is ranked by the individual out of 10.

Which of the following values for n, f, g (respectively) makes the most sense?

1. 0, 8, 3
2. 10, 5, 8
3. 0, 0, 8
4. 10, 0, 8
5. 0, 5, 8
6. 10, 8, 3

20. In machine evolution, we have 4 individuals, where individual i has fitness $f(i)$:

i	1	2	3	4
$f(i)$	0	5	4	5

If we use a "Soft max" probability distribution with a "temperature" parameter T to choose who reproduces, then as $T \rightarrow \text{infinity}$ the probability of each individual reproducing goes to:

1.

$1/4$	$1/4$	$1/4$	$1/4$
-------	-------	-------	-------

2.

0	$5/14$	$4/14$	$5/14$
---	--------	--------	--------

3.

0	$1/2$	0	$1/2$
---	-------	---	-------

4.

0	0.9	0.1	0.9
---	-----	-----	-----

5.

0	1	0	1
---	---	---	---

6.

0	1	0	0
---	---	---	---

Long questions

Question 1

[Total marks: 25]

- 1(a) "A better AI would be one that nailed down the definition of intelligence first, before proceeding." Discuss. [7 marks]
- 1(b) The Turing Test proposes a test for intelligence where, if you cannot tell the difference between the machine's responses and human responses, you might propose that the machine actually is intelligent. I wrote a chatbot that passed the Turing Test in 1989 and I do not think it is intelligent. How can this be? [6 marks]
- 1(c) AI people love competitions. Why? And can you think of a way in which competitions can be unhealthy for AI research? [6 marks]
- 1(d) A chess playing machine examines millions of moves per second. It is common to say this is not how humans play. Or is it? [6 marks]

Question 2

[Total marks: 25]

2(a) Consider a heuristic search as follows:

[13 marks]

```
Initialise with 20 random solutions.
Repeat for a long time:
{
  Test the 20 solutions.
  Pick the best 10.
  Make small changes in them
  to get 20 new solutions.
}
```

- i) Explain what this heuristic is trying to do. [2 marks]
- ii) Explain how this heuristic may fail. [2 marks]
- iii) What would happen if you excluded the last step in the loop? ("Make small changes in them") [3 marks]
- iv) What would "Pick the best 1" look like? [3 marks]
- v) What would "Pick the best 18" look like? [3 marks]

2(b) i) Explain "Algorithm A".

[12 marks, 4

ii) Explain how "Algorithm A" compensates for an *inaccurate* heuristic evaluation function.

each]

iii) Are heuristic evaluation functions often inaccurate?

Question 3

[Total marks: 25]

3(a) A mobile robot has Input state = (angle of direction of nearest wall, velocity value). Output behaviour = one of (turn left, turn right, stop, keep going).

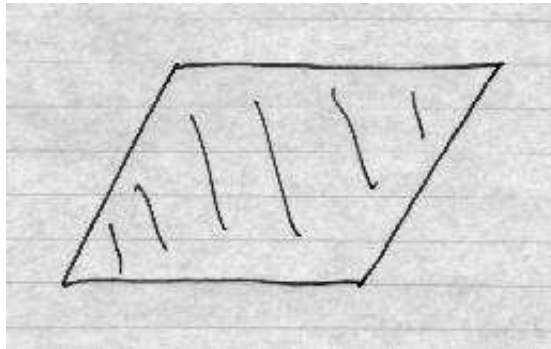
[5 marks]

What information is the Input state missing? What are the consequences of this missing information?

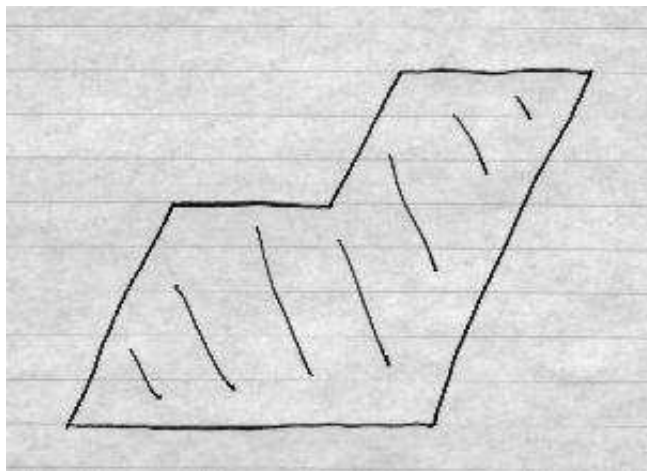
3(b) In a multi-layer network:

[8 marks,
4 each]

- i) Explain how you can make a multi-layer network that fires if and only if the input point is inside the parallelogram in this figure:



- ii) Explain how you can make a multi-layer network that fires if and only if the input point is inside this figure:



3(c) Consider crossing over two parents to produce two offspring.

[12 marks, 4
each]

- i) Under what circumstances will the crossover operation *not* produce any new diversity?
- ii) Show how crossover only at gene boundaries produces new diversity but no new genes.
- iii) Show how crossover at random locations produces new genes.

Bonus question (for 10 marks)

*This question is to make you do some reading.
You can prepare this question in advance.*

Tell me about some AI-related algorithm or software that was *not* mentioned on the course.

CA318 - Advanced Algorithms and AI Search

Answer sheet

Black out the square that matches the correct answer.

Exam No: _____

Class: _____

Seat No: _____

Question	1.	2.	3.	4.	5.	6.	7.	8.
1								
2								
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Question	1.	2.	3.	4.	5.	6.	7.	8.
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